

Assignment 4 Report

Repository Links

- GitHub Repository: <https://github.com/NisargUpadhyayIITJ/MLOps-NisargUpadhyay-B23CS1075>
- HF Repository: <https://huggingface.co/NisargUpadhyay/b23cs1075-assignment4-models>

Baseline Metrics

- Training Time (100 epochs): 52.32 minutes
- Final Loss: 0.0974
- BLEU Score (after 100 epochs): 52.47

Hyperparameters Tuned

Hyperparameter	Search Range
learning_rate	loguniform(1e-5, 1e-3)
batch_size	choice([16, 32, 64])
num_heads	choice([4, 8])
d_ff	choice([1024, 1536, 2048])
dropout	uniform(0.10, 0.40)
num_layers	choice([4, 6])
weight_decay	loguniform(1e-6, 1e-3)

Best Configuration Found

- Configuration: {"learning_rate": 8.884411309500873e-05, "batch_size": 16, "num_heads": 4, "d_ff": 1536, "dropout": 0.19016462895143443, "num_layers": 4, "weight_decay": 1.149540832915101e-06, "d_model": 512, "max_len": 50, "seed": 42, "force_cpu": false, "epochs": 20}
- Sweep size used for current run: 20 trials x 20 epochs
- Best trial BLEU: 76.98
- Best trial loss: 0.5256

Final Metrics of Best Model

- Training Time: 14.22 minutes
- Final Loss: 0.5256
- BLEU Score: 84.72
- Epochs required to match/beat baseline: 6

Notes

- Baseline reference source: Extracted from en_to_hi.ipynb recorded outputs on 2026-03-13

- Ray Tune reports both `loss` and `bleu` each epoch via `tune.report(...)` for compatibility with the installed Ray version.
- ASHA stops weak trials early to reduce compute cost during the Optuna-guided sweep.